

ELOWEN COX, PE, PMP

ENGINEERING & INFRASTRUCTURE EXECUTIVE

Roanoke, Virginia | (540) 558-8986 | elowen.s.cox@gmail.com | www.linkedin.com/in/elowen-cox

Water & Healthcare Infrastructure | Capital Programs | Enterprise Risk | Federal Operations | Research & Innovation

EXECUTIVE PROFILE

Engineering and infrastructure leader with 15+ years of progressive experience across federal infrastructure, healthcare engineering, capital program delivery, water quality, military command, and applied research. Demonstrated ability to lead large multidisciplinary organizations, oversee multimillion-dollar capital portfolios, advise senior leaders on complex infrastructure and public-health risks, and translate technical issues into executable strategies. National-level subject matter expertise in water quality and opportunistic pathogens complements extensive owner-side project and facilities leadership.

LEADERSHIP SCALE & SELECTED RESULTS

- Led a 90-person multidisciplinary VA engineering organization supporting a 1.2M-GSF, 270-acre medical campus.
- Oversaw a VA capital construction program exceeding \$65M, including a new utility plant and mixed oxidant injection system for Legionella mitigation.
- Managed more than \$31M in complex GSA projects involving federal buildings, historic structures, a seawall, warehouse construction, and major building systems.
- Simultaneously managed five USACE construction projects valued at more than \$21M.
- Identified contract-change errors that saved the Government 50 days and \$67,000; developed a quality-control system supporting on-time turnover of 110 housing units.
- Commanded a 96-person Army engineer organization with accountability for more than \$40M in government property and trained 60+ soldiers supporting a 3,500-person brigade combat team.

EXECUTIVE LEADERSHIP COMPETENCIES

Enterprise Engineering Leadership • Capital Strategy & Delivery • Infrastructure Risk & Resilience • Owner Advisory • Healthcare Infrastructure • Water Quality • Opportunistic Pathogens • Portfolio Management • Strategic Planning • Budget Development & Tracking • Construction & Design • Federal Acquisition • Organizational Leadership • Talent Development • Regulatory Coordination • Technical Due Diligence • Executive Communication • Research & Innovation

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | Virginia Tech, Department of Civil & Environmental Engineering

July 2025 – Present | Blacksburg, VA

- Conduct independent research under the supervision of two Primary Investigators in a BSL-2 laboratory.
- Design experiments, maintain data integrity, mentor undergraduate research assistants, and develop novel approaches to mitigate opportunistic pathogens in premise plumbing systems.
- Apply pH and dissolved oxygen measurements, flow cytometry for total and intact cell counts, Legioalert for Legionella pneumophila quantification, and direct digital PCR targeting specific genes.

Healthcare Engineer, Civil/Environmental | U.S. Department of Veterans Affairs, VHA Central Office

September 2023 – September 2025 | Washington, DC / Remote

- Served as VHA subject matter expert on water quality and opportunistic pathogens in potable and process water; provided consultative services across the VHA enterprise.
- Advised organizational levels of VA, external contractors including Booz Allen Hamilton, and industry partners on trends and research in waterborne pathogens.
- Created training materials and delivered training on AAMI ST 108 to national engineering groups, OSP, and VISNs; modified approaches used by facilities to install water-treatment systems.
- Drafted primary responses for IPT concurrence on AAMI ST 108 and coordinated with OHE and OSP partners to efficiently provide field guidance.
- Coordinated regional, national, and department-level stakeholders implementing policies and program plans associated with VHA Directive 1061.
- Conducted cause-based site visits at Loma Linda and Buffalo VA medical centers; analyzed bacterial proliferation and reviewed water-distribution, elevated-storage, and stormwater designs.

Associate Chief Engineer | U.S. Department of Veterans Affairs, Salem VA Medical Center

December 2018 – September 2023 | Salem, VA

- Led a 90-person Engineering, Biomedical, Maintenance, and Utility Plant organization supporting a 1.2M-GSF clinical and administrative campus and 270-acre grounds.
- Managed facilities management, construction, strategic planning, budgeting, utility operations, repair, renovation, modernization, and Biomedical Engineering program oversight.
- Executed a capital construction program exceeding \$65M, including a new utility plant, mixed oxidant injection system for Legionella mitigation, air-handler replacement, and transformer replacement.
- Implemented the Medical Center Director's Strategic Plan by programming projects in the Strategic Capital Investment Planning tool and integrating facility needs and risks into a comprehensive healthcare infrastructure plan.
- Managed the environment-of-care and life-safety program and successfully supported Joint Commission reaccreditation at the beginning of the COVID-19 pandemic.

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- Evaluated stormwater, roads, clinical and administrative systems, water distribution, and electrical infrastructure; prioritized repair and replacement using engineering judgment.
- Implemented Legionella mitigation strategies and coordinated with the Virginia Department of Health regarding compliance for a mixed oxidant injection system on the potable water supply.

Project Manager / Civil Engineer | U.S. General Services Administration, Design & Construction

October 2016 – December 2018 | Boston, MA

- Managed diverse federal projects exceeding \$31M, including fire-alarm and building-automation replacement in an 11-story federal building, warehouse design/construction, historic Custom House masonry restoration, 100-year-old seawall replacement, and federal courthouse window replacement.
- Wrote complete bid packages including scopes, specifications, and drafted drawings for historical restoration work.
- Monitored and controlled construction budgets; coordinated technical review of change orders with A/E teams to ensure best value for the Government.
- Wrote statements of work for a Program Development Study and a Design-Build Army warehouse as part of a real-property exchange agreement.

Project Engineer | U.S. Army Corps of Engineers, Fort Wainwright

July 2015 – September 2016 | Alaska

- Simultaneously managed five diverse construction projects valued at more than \$21M, including contaminated-soil remediation, building rehabilitation, earthwork/site preparation, and residential housing.
- Coordinated multidisciplinary project delivery teams from planning charrettes through completion to keep projects on schedule and within budget.
- Reviewed contractor schedules using Primavera P6; identified errors in a contract change that saved the Government 50 days and \$67,000.
- Developed a quality-control support system that enabled rapid issue identification and resulted in on-time turnover of 110 housing units.

Engineer Company Commander / Project Manager | U.S. Army

May 2014 – June 2015

- Commanded a 96-person engineer organization; maintained direct responsibility for more than \$40M in government property and led training, safety, personnel actions, and morale.
- Served as technical advisor to the commander, providing engineering assessments and solutions for expeditionary route construction and site improvement in remote Alaska and California.
- Developed organizational expectations for five officers and one senior non-commissioned officer and was recognized by senior command for an exceptional organizational climate based on mutual trust and respect.
- Trained more than 60 soldiers on military explosives and planned, resourced, and executed certification events enabling support to a 3,500-person brigade combat team.

RESEARCH & TECHNICAL EXPERTISE

Potable water distribution • Premise plumbing • Water quality • Legionella pneumophila • Opportunistic pathogens • Biofilms • Microbial ecology • Flow cytometry • Total/intact cell counts • Direct digital PCR • FISH • Molecular detection • Genomics • Metagenomics • Microbial transport • Infrastructure risk • Water treatment • Healthcare engineering • Capital planning • Construction management

EDUCATION

VIRGINIA TECH — PhD, Civil & Environmental Engineering, enrolled. Research focus: opportunistic pathogens in potable water distribution and premise plumbing.

MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY — M.S., Engineering Management, 2013. GPA 4.0/4.0.

VIRGINIA MILITARY INSTITUTE — B.S., Civil & Environmental Engineering, 2008. GPA 3.84/4.0; CE GPA 3.95/4.0.

LICENSES & CREDENTIALS

Professional Engineer, Virginia (Current) • Project Management Professional (Current)